

Statistical Interpretation and Exploratory Data Analysis of the Human Development Index (HDI)

Student Name: Prajita Banjara

Student ID: 2513644

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Biratnagar International College

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Synopsis

A popular composite indicator, the Human Development Index (HDI) gauges a nation's overall progress by taking into account factors like income, education, and health. With an emphasis on global trends, South Asian development patterns, and a comparison between South Asia and the Middle East, this study conducts statistical interpretation and exploratory data analysis (EDA) of HDI data. Several visualizations and statistical methods are used to find trends, disparities, correlations, and outliers using data from 2020 to 2022. Significant regional disparities, the influence of income and health on development outcomes, and the significance of inclusive growth that goes beyond economic performance are all highlighted by the findings.

1. Description

Human development includes advancements in living standards, health, and education in addition to economic growth. The United Nations Development Program (UNDP) created the Human Development Index (HDI), which offers a thorough metric for comparing development results between nations and regions.

This report's goal is to use statistical interpretation and exploratory data analysis methods to examine HDI data. The analysis is centered on:

Recognizing HDI trends in South Asian nations,

Finding connections between HDI and its essential elements,

Identifying development gaps and anomalies, and contrasting the Middle East's and South Asia's development outcomes.

The study employs visualization techniques to extract significant insights from cleaned and processed HDI data from 2020 to 2022.

2. Information and Techniques

The Human Development Index Dataset, which spans several nations and years, was the dataset utilized for this analysis. The HDI, Life Expectancy Index, Gender Development Index, and Gross National Income (GNI) per capita are important factors.

Preprocessing of the data involved:

Data filtering for particular years (2020–2022),

Making regional divisions for the Middle East and South Asia,

Managing absent values and guaranteeing uniform data types,

producing derived metrics like the GNI–HDI Gap and Composite Development Score.

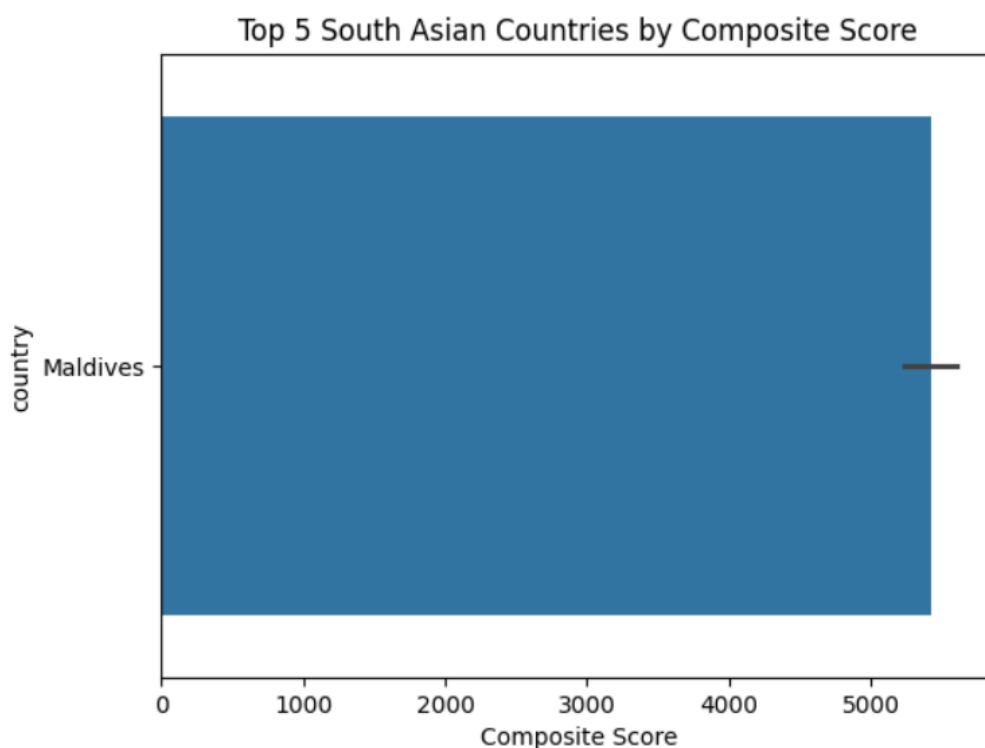
Bar charts, scatter plots with trend lines, and the $1.5 \times \text{IQR}$ rule for outlier detection were among the exploratory data analysis methods.

3. Exploration of Advanced HDI: South Asia

3.1 Total Development Points

To offer an alternative viewpoint on development performance, a Composite Development Score was created using weighted contributions from GNI per capita and the Life Expectancy Index.

The top five South Asian nations by Composite Score are shown in Figure 1. The combined impact of income and longevity on development is demonstrated by the higher ranking of nations with greater economic capacity and improved health outcomes. The composite score emphasizes the disparities in South Asian countries' development priorities, whereas the HDI captures broader dimensions.



The top-ranked South Asian nation according to Composite Development Score is shown in Figure 1.

3.2 Outlier Detection: HDI vs. GNI per Capita

The outliers were found using the criterion of $1.5 \times \text{IQR}$ in order to determine deviations from expected development pathways.

In Figure 2 below is a scatter diagram showing HDI and per-capita GNI. Some countries have demonstrated strong income without necessarily showing strong HDI. This is evident in countries that have weaknesses in structural inequality or investments made in education and health. Some countries with low income yet strong HDI have invested well despite limited income.

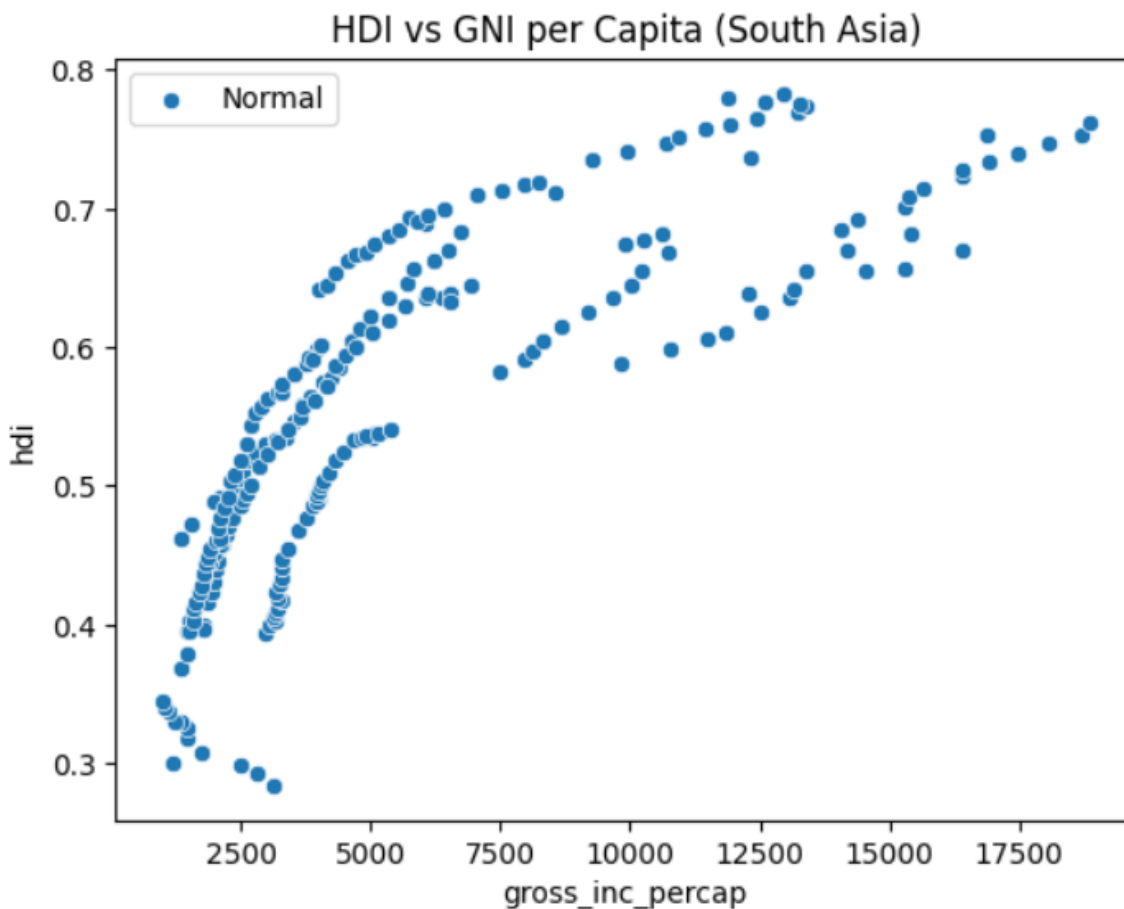
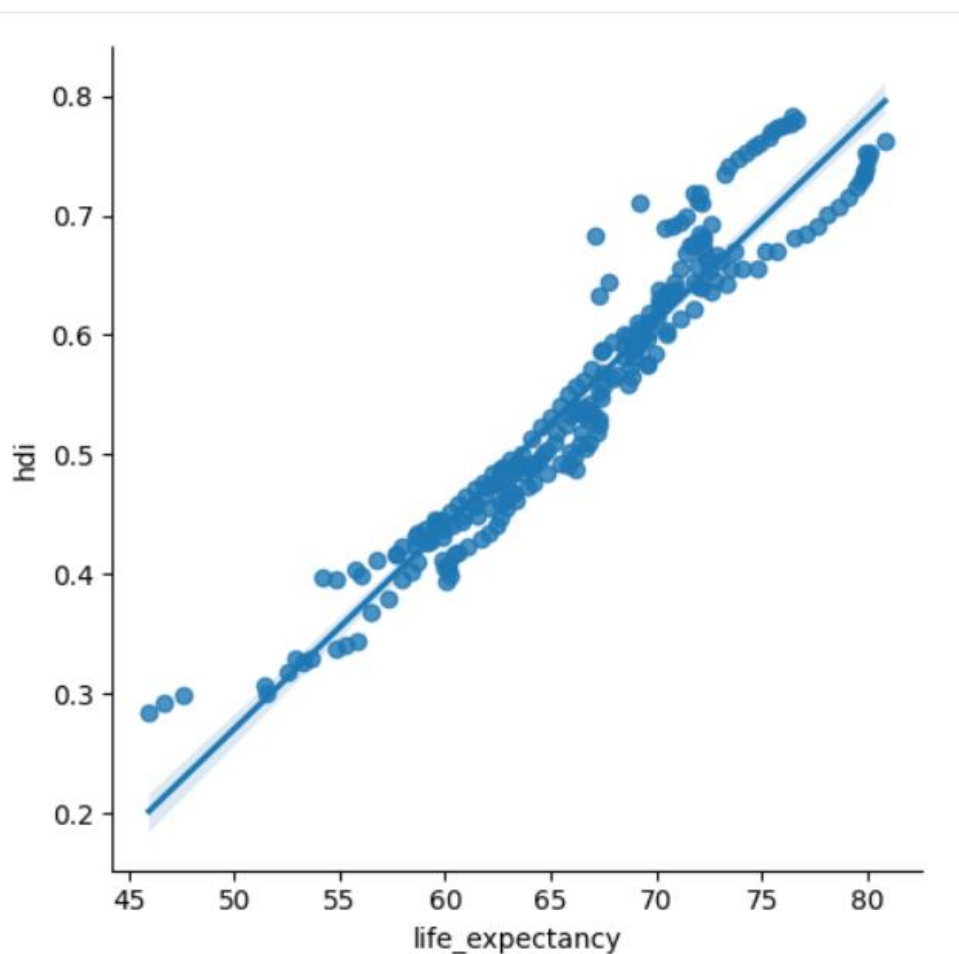


Figure 2 shows a scatter plot of GNI per capita versus HDI with outliers highlighted.

3.3 The Connection Between Development Indicators and the HDI

Correlation analysis was used to determine how strongly the HDI and its constituent parts were related.

Scatter plots with trend lines for the HDI against the Gender Development Index and Life Expectancy Index are displayed in Figure 3. Life expectancy and HDI have a strong positive correlation, demonstrating the vital role that health outcomes play in human development. Gender development and HDI have a positive, albeit weaker, relationship.



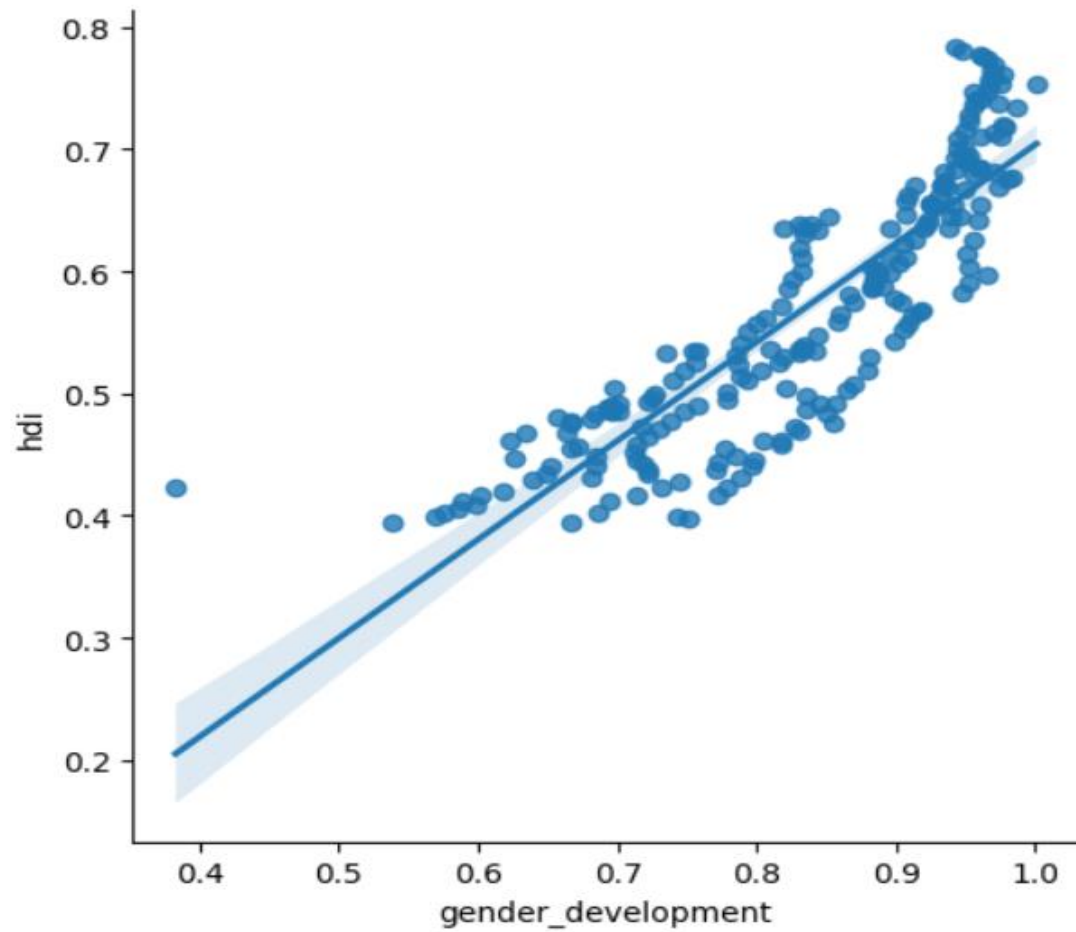
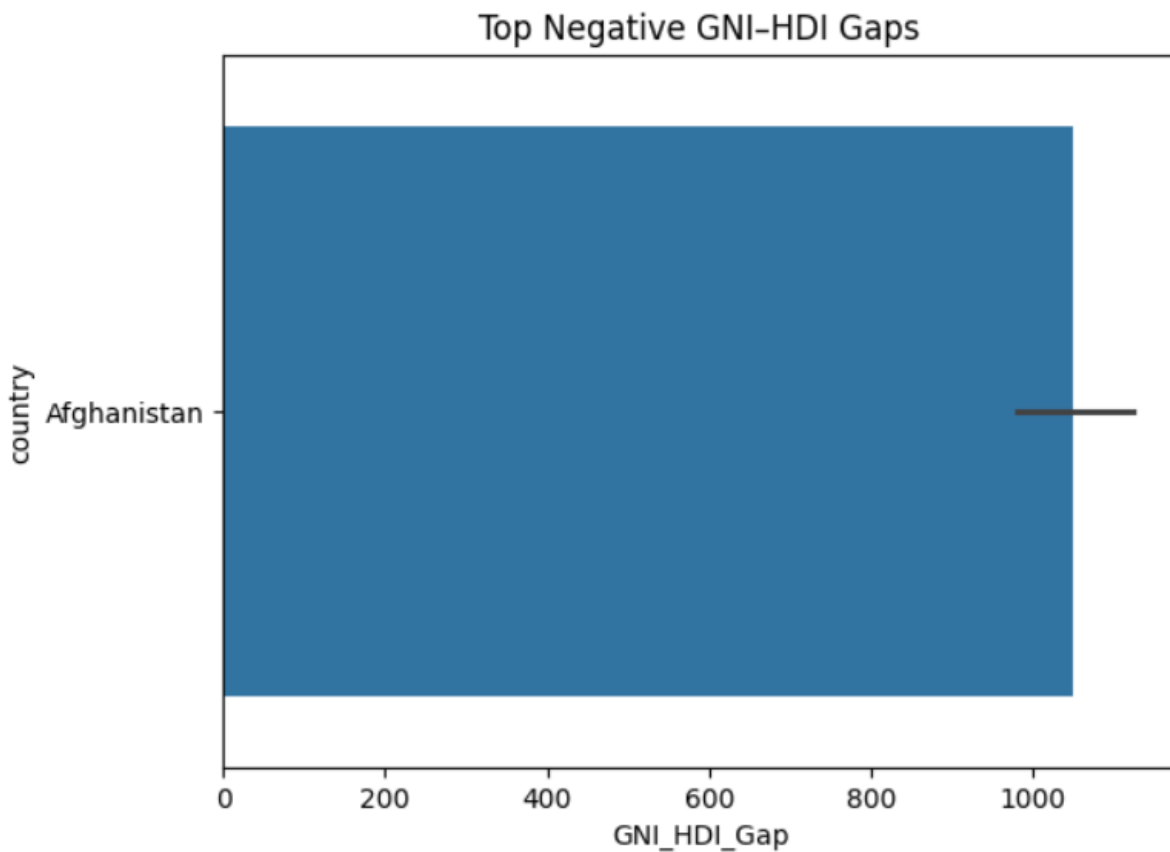


Figure 3 shows the relationship between the Gender Development Index, Life Expectancy Index, and HDI.

3.4 Analysis of the GNI-HDI Gap

To find discrepancies between income levels and human development outcomes, the GNI–HDI gap was computed.

The top three positive and negative GNI–HDI gaps are shown in bar charts in Figure 4. Negative gaps show countries with relatively high HDI despite lower income levels, while positive gaps show countries where income growth has not translated into comparable human development. This highlights the fact that comprehensive development requires more than just economic growth.



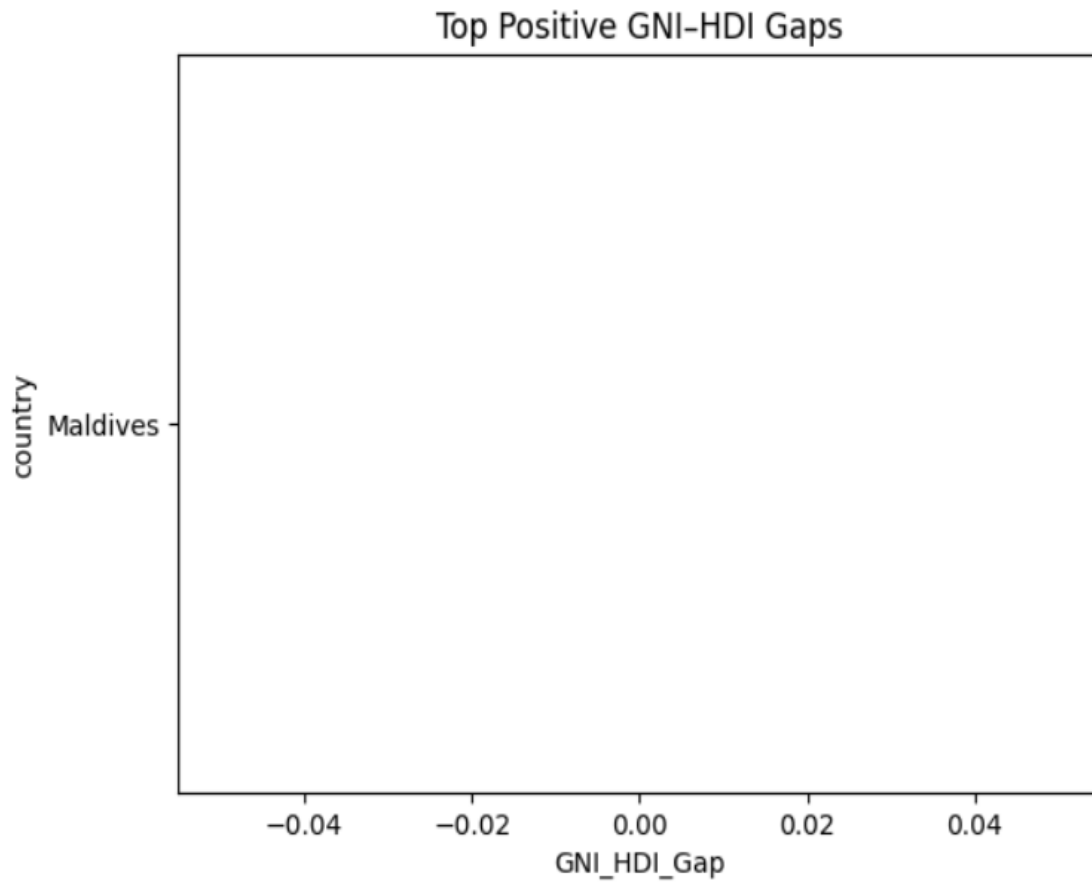


Figure 4 shows South Asia's top GNI-HDI gaps, both positive and negative.

4. Regional Comparison: Middle East vs. South Asia

4.1 High and Low Achievers

HDI data from 2020 to 2022 were used to perform a regional comparison.

The top three and bottom three HDI-performing nations from the Middle East and South Asia are contrasted in Figure 5. Significant regional differences are reflected in the fact that South Asian countries are more frequently found among the lower performers, while Middle Eastern countries typically hold the top spots.

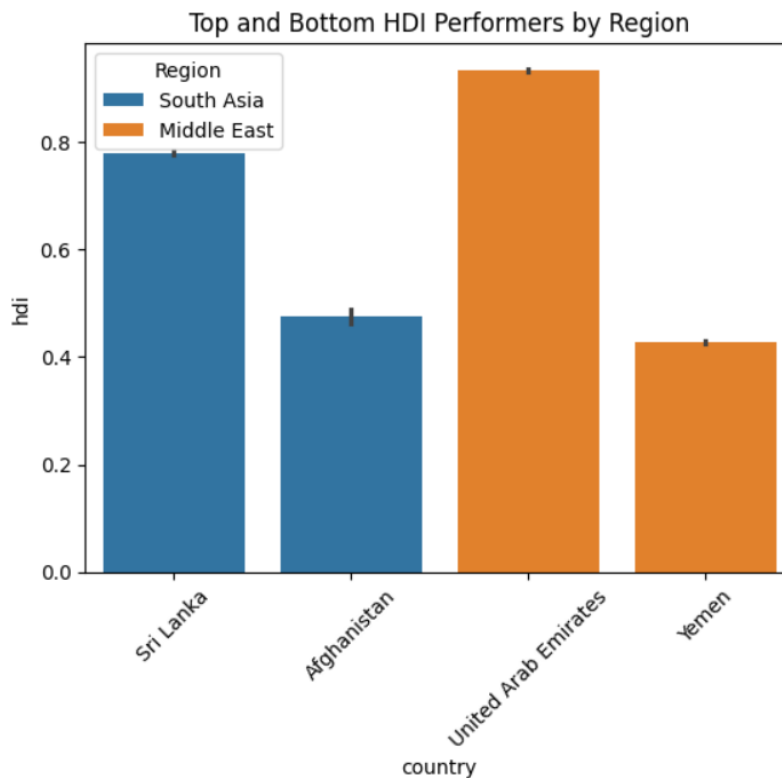


Figure 5 shows the top and bottom HDI performers for the Middle East and South Asia.

4.2 Major Improvement Dimensions Evaluation

To identify the causes of disparity, key development indicators were compared between regions.

The Gender Progress Index, Life Expectancy Index, and GNI per capita are shown as grouped bar charts in Figure 6. The biggest difference among these is in GNI per capita, with Middle Eastern nations outperforming South Asian nations by a wide margin. While gender development exhibits comparatively less variation, life expectancy differences are also noteworthy.

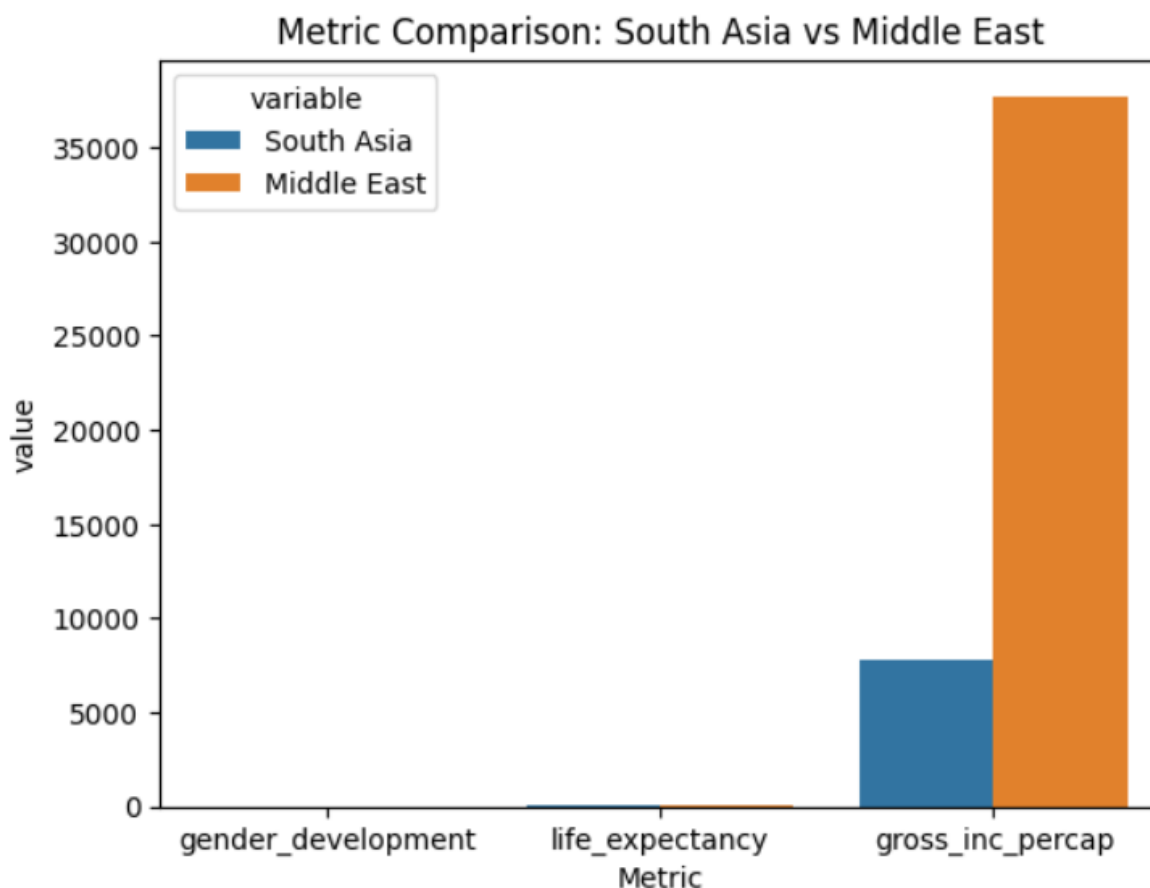
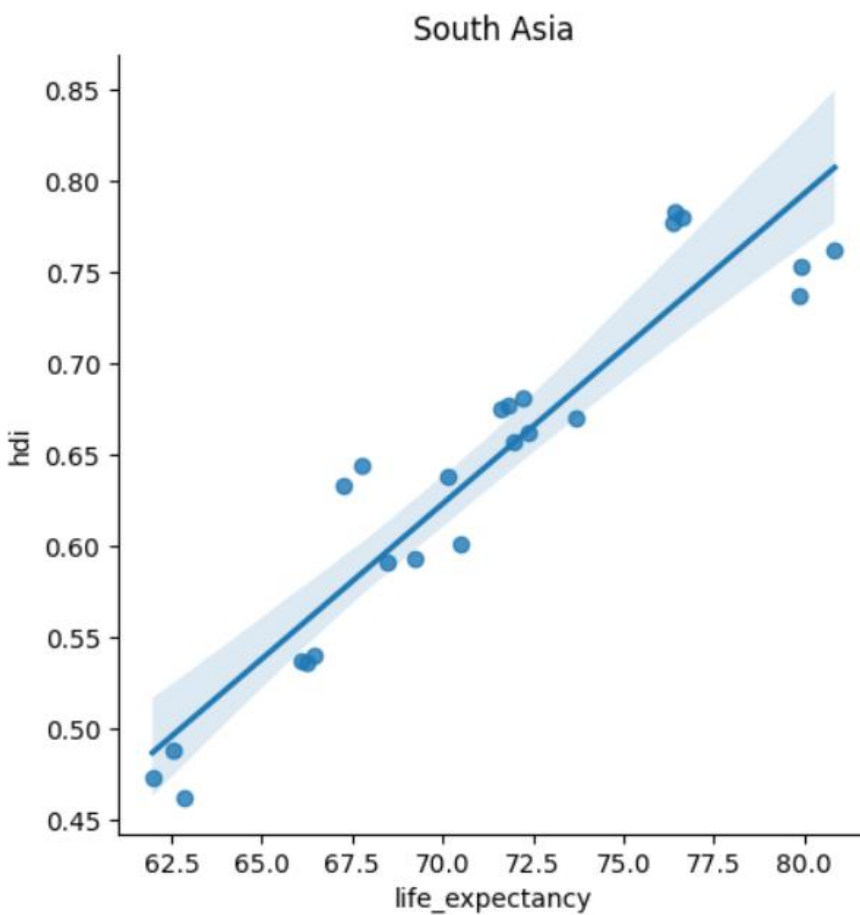


Figure 6 shows a comparison of Middle Eastern and South Asian development metrics.

4.3 Geographic correlation study

The association between HDI and life expectancy in both regions was investigated using correlation analysis.

Scatter plots with trend lines for the Middle East and South Asia are shown in Figure 7. Life expectancy and HDI show a strong positive correlation in both regions, but the relationship is steeper in South Asia, indicating that improvements in health result in higher HDI gains there.



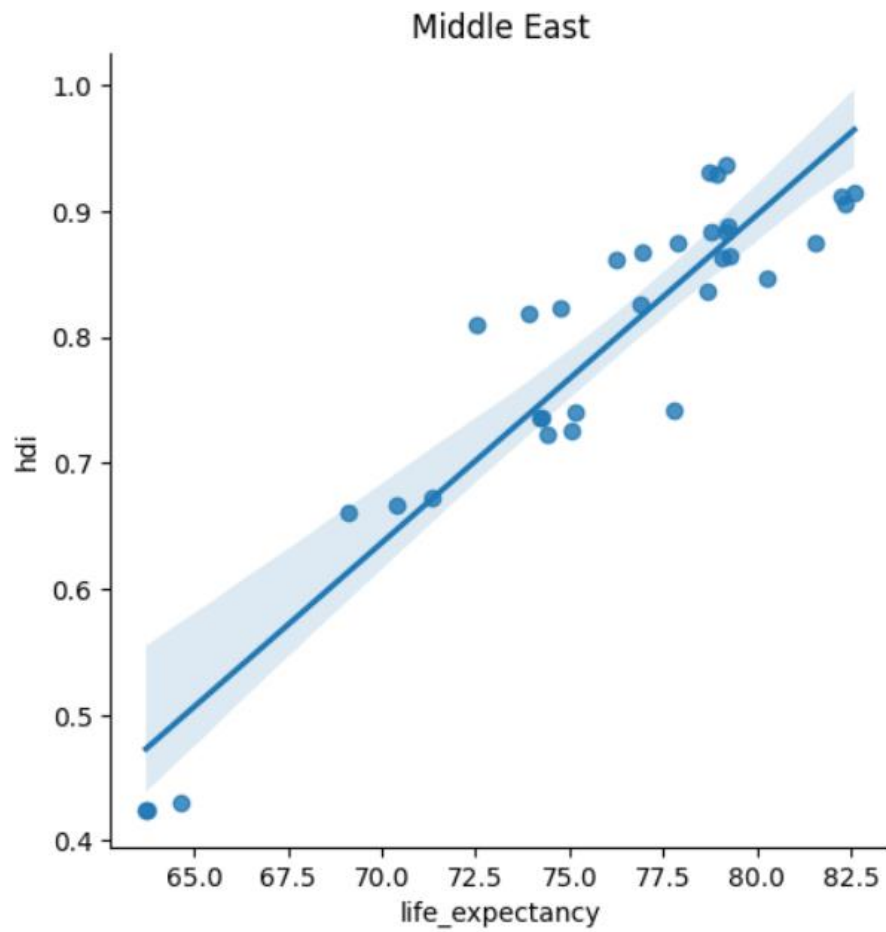


Figure 7: Life Expectancy by Region versus HDI.

5. Final verdict

Significant disparities in human development exist both within and between South Asian regions, according to this exploratory data analysis. Health and social factors are just as important in determining HDI as income. The existence of outliers and development gaps emphasizes the significance of inclusive growth and successful social policies. Overall, the results show that balanced advancements in the social, health, and economic spheres are necessary for sustainable human development.

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